Rohan Vasireddy

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EDUCATION

University of California, Berkeley **Berkeley, CA**
Bachelor of Arts in Economics & Data Science *Exp. Graduation: May 2030*
Relevant Coursework: Linear Algebra, Microeconomics, Macroeconomics, Introduction to Probability and Statistics

EXPERIENCE

Software Engineer Intern **Riverside, CA**
Docere *08/2025 – Present*

- Built core application features for an AI-powered coding environment serving 1K+ students at a venture-backed EdTech startup, supporting browser-based programming workflows across the platform.
- Engineered a COPPA-compliant, multi-language execution sandbox for Python, JavaScript, and HTML, using isolated runtimes to safely execute student code while separating workloads from core application services.
- Strengthened platform reliability, observability, and request security by implementing centralized asynchronous error handling, request-level Sentry monitoring, and CSRF protection across application APIs.

Research Intern **Riverside, CA**
University of California, Riverside *08/2025 – Present*

- Developed MATLAB computational models to simulate hydrodynamic and elastic interactions in viscous fluids, quantifying bead displacement behavior on soft elastic surfaces.
- Applied statistical regression and signal processing to optical microscopy data across 15+ experimental conditions, generating reproducible, peer-reviewable datasets.
- Selected from 200+ applicants to present research at the APS 77th Annual Division of Fluid Dynamics Meeting in Salt Lake City.

Investment Analyst **Riverside, CA**
Lancer Financial Collective *08/2024 – 07/2026*

- Built equity screening models in Excel using DCF valuation and P/E ratio benchmarking to evaluate 40+ securities across 8+ sectors, identifying undervalued positions and reducing portfolio risk.
- Delivered data-driven investment recommendations to fund managers using structured reports, directly influencing allocation decisions for a $200K university-managed portfolio.
- Conducted risk-adjusted return analysis using Sharpe ratios and beta coefficients to guide quarterly rebalancing and align sector exposure with macroeconomic trends.

Assistant Center Director **Fontana, CA**
Mathnasium of South Fontana *06/2024 – 05/2026*

- Managed daily center operations including scheduling 100+ weekly sessions, handling enrollment inquiries, and serving as primary point of contact for academic concerns.
- Maintained KPI dashboards tracking 10+ metrics — retention rate, attendance, and enrollment conversion as well as delivering weekly performance reports to management.
- Processed 50+ monthly accounts payable/receivable transactions; supported payroll reconciliation, invoice management, and budget variance analysis.

PROJECTS

US Economic Indicator Dashboard

- Built a reproducible data pipeline (Python, R, SQL) fetching 10 years of U.S. macroeconomic data from the FRED API, loading into a SQLite database with normalized tables and 3 analytical views
- Ran multivariate regression (R^2=0.41) and correlation analysis across 5 indicators; automated 5 ggplot2 charts and a formatted 4-sheet Excel workbook as reproducible outputs

SKILLS & CERTIFICATIONS

Technical: Python (Pandas, NumPy, Matplotlib), Java, SQL, R, MATLAB, Excel (PivotTables, Power Query, VLOOKUP), Statistical Modeling, Data Visualization

Financial Skills : Valuation, Financial Modeling, Portfolio Analysis, Risk Analysis

Frameworks: Scikit-learn, TensorFlow, Pandas, OpenCV, React, REST APIs

Tools: Tableau, Power BI, Jupyter Notebook, GitHub, Google Colab

Certifications: AWS AI & ML Scholars, Mckinsey Forward Scholar, IBM Product Manager